國立成功大學 - 機械工程學系

機械專題實作

Topic: 工程設計程序 & 歷屆專題實作回顧

劉至行

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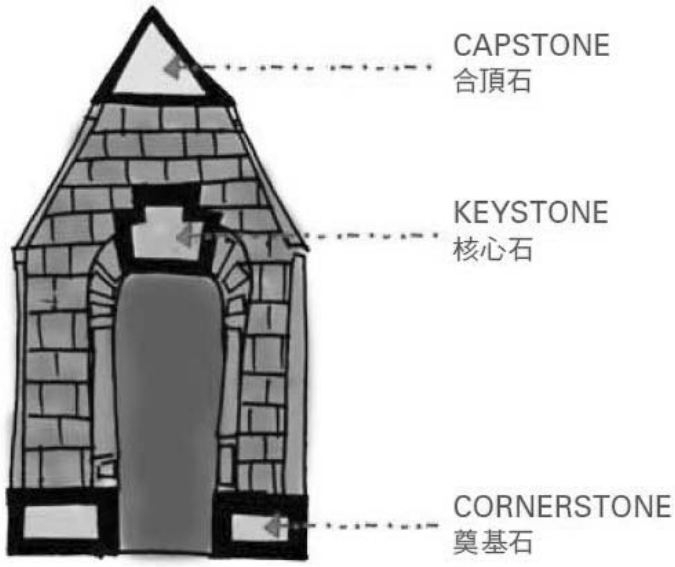
Date: Feb 2026

Contents

- Capstone Design
- Engineering Design Process
- Project Management
- Literature Review
- 歷屆機械專題實作回顧

Capstone（合頂石）指的是建築物最頂端、最後一塊石頭，用以穩固建築結構，使其順利完工。除了 capstone 之外，尚有 keystone（核心石）以及 cornerstone（奠基石），同樣也在建築結構中扮演相當重要的角色。Keystone 為拱門頂端的楔形石頭，用以使整體結構緊密銜接。Cornerstone 則為建築物底部角落的柱石，用以支撐整體建築結構。



[Source: 台大教學發展中心]

The Tower of Knowledge
知識之塔

由三本書為意象所構成的金字塔。由下而上分別代表不同階段的課程：基礎課程 (Cornerstone course)、核心課程 (Keystone course) 以及總整課程 (Capstone course)。

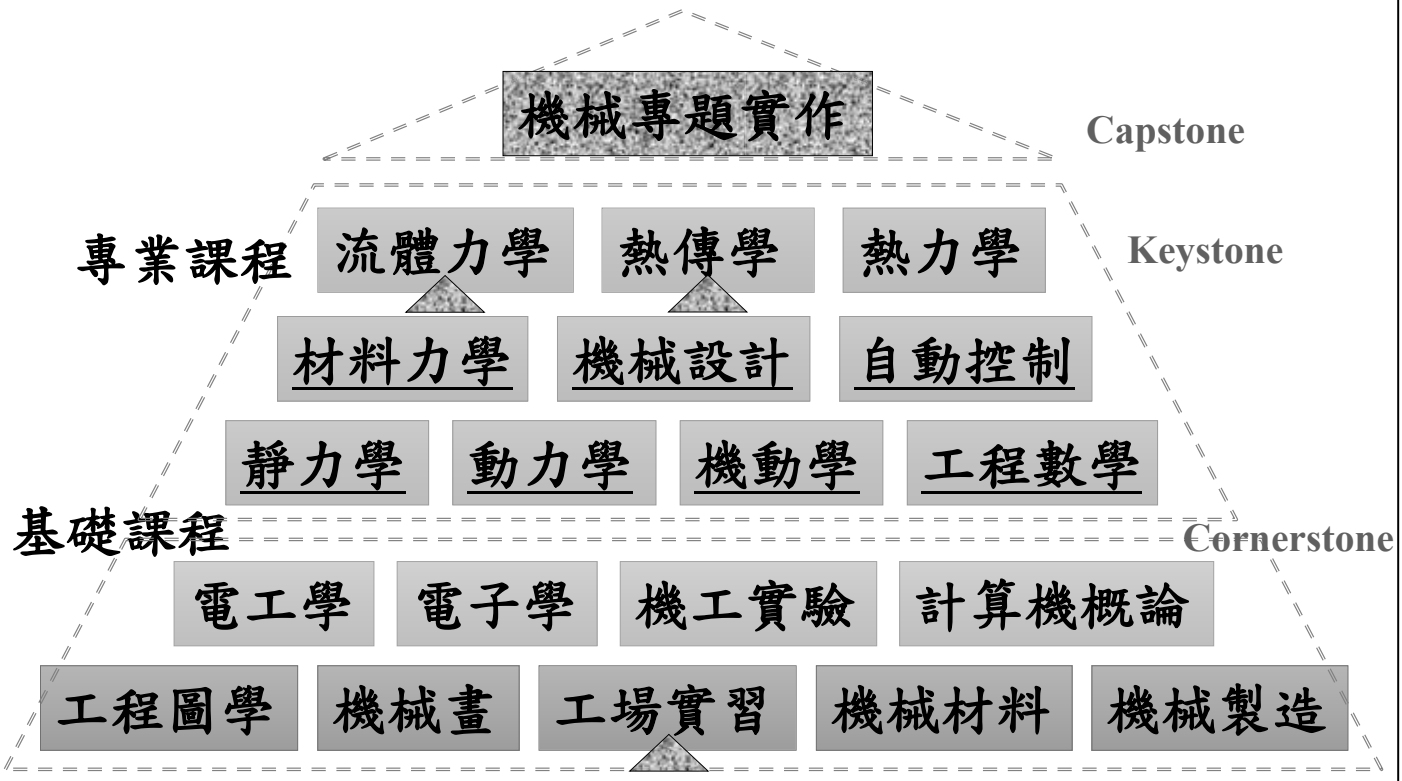
基礎課程
Cornerstone course
奠基石用以支撐整體建築結構。為共同或領域基礎課程，支撐未來學習。

核心課程
Keystone course
核心石用以緊密銜接建築結構。專業領域的核心課程，緊密銜接專業學習。

總整課程
Capstone course
合頂石用以穩固整體建築結構。統整深化大學所學，穩固完成所有學習。

[Source: 台大教學發展中心]

成大機械系必修課程規劃設計

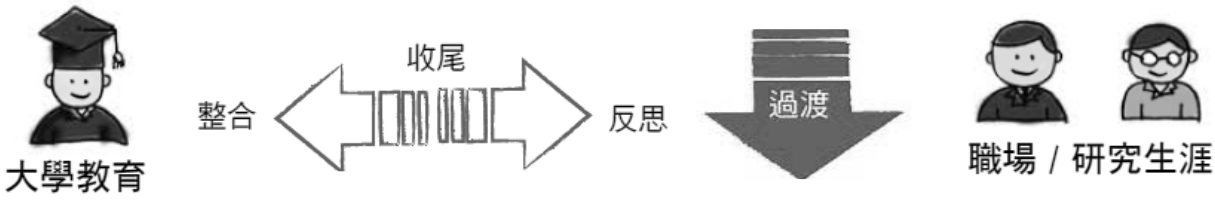


(1) 整合 (integration)：促使學生總整大學所學，讓大學學習並非只是一塊塊獨立的磚頭，而是完整的房子；並非只是學分的累積，而是有意義的整體學習經驗。

(2) 收尾 (closure)：為大學經驗畫下句點。

(3) 反思 (reflection)：提供機會讓學生反省思考大學四年學到了什麼? 還有哪些不足?

(4) 過渡 (transition)：協助學生順利連結大學經驗以及畢業後生涯，為未來做好準備。



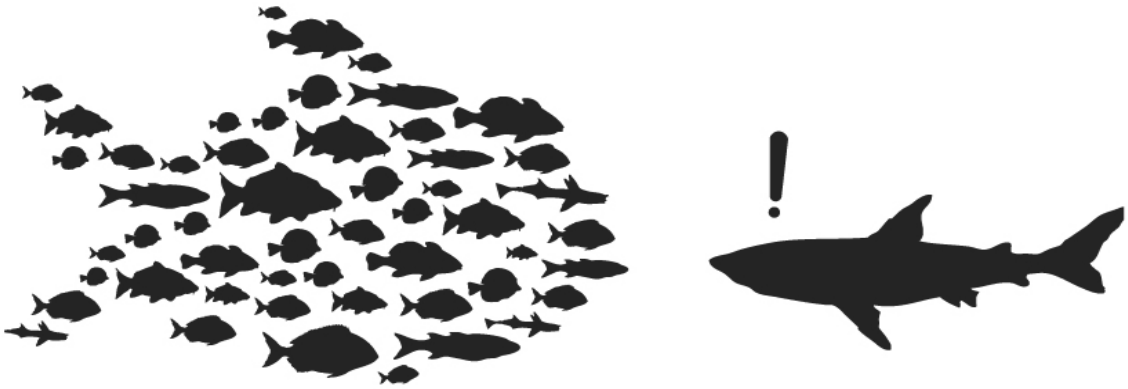
[Source: 台大教學發展中心]

https://www.jsae.or.jp/formula/en/about.php

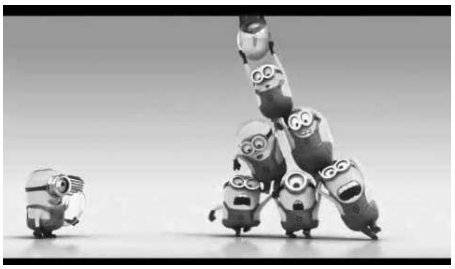
Fundamental Policies of the Competition

- as an engineering society, to provide students with an opportunity for monozukuri for the purpose of:
  - (1) helping the students to **independently** develop their comprehensive monozukuri skills.
  - (2) increasing the educational value of the experience by providing the students with an opportunity for the practical application of skills and knowledge that are connected to their classroom studies.

Monozukuri或拼寫Monodzukuri，在日語中字面意思是“生產”或“製造事物”，是日語中“製造”的術語。更廣泛的含義包括日本製造業實踐的技術實力，專有技術和精神的綜合。這種精神包括以自豪，技能和奉獻精神對生產的真誠態度，以及對創新和完美的追求。 维基百科（英文)

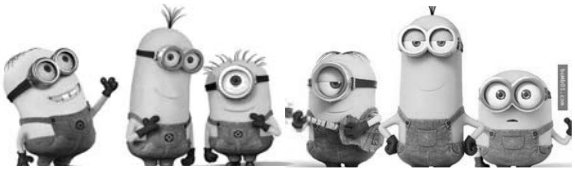


TEAMWORK

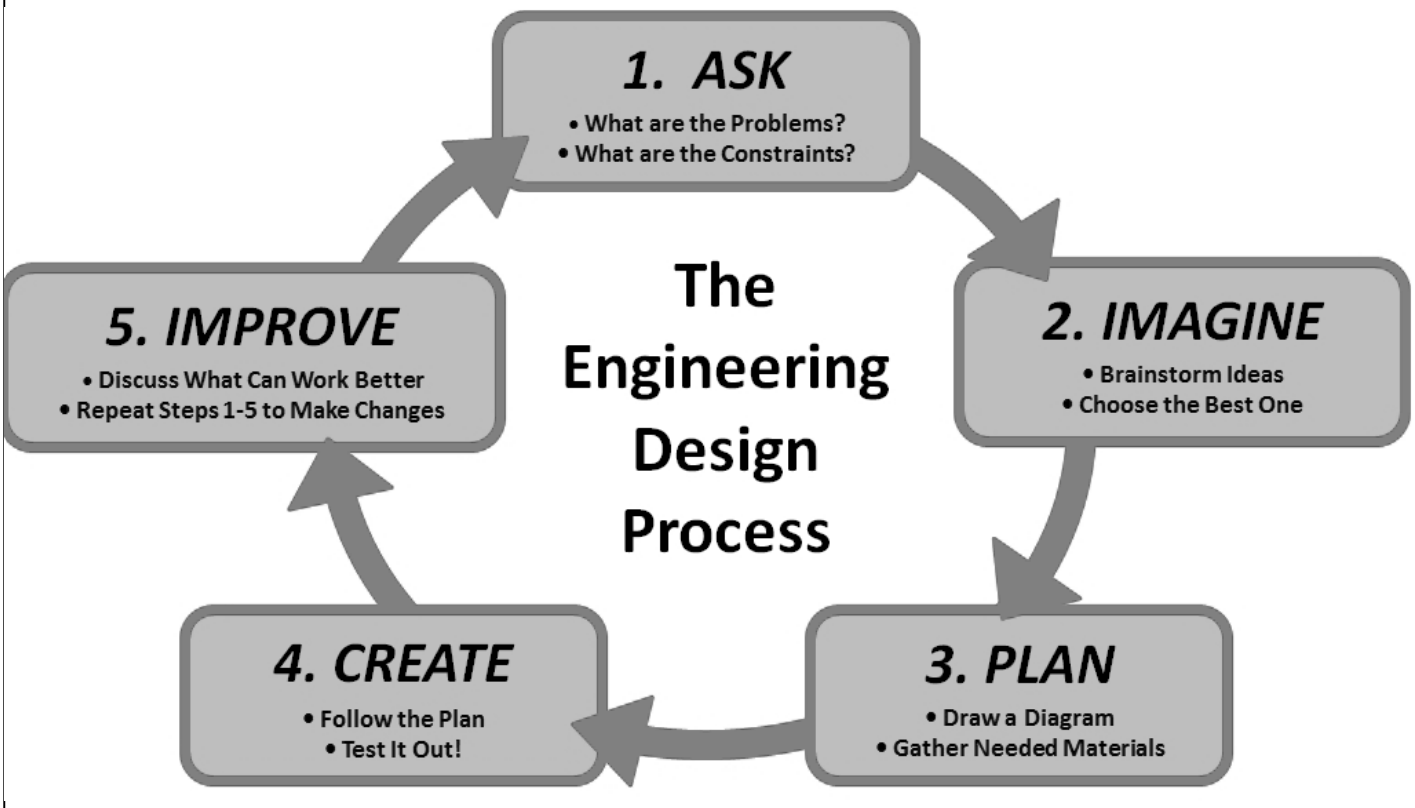


searchenginepeople.com

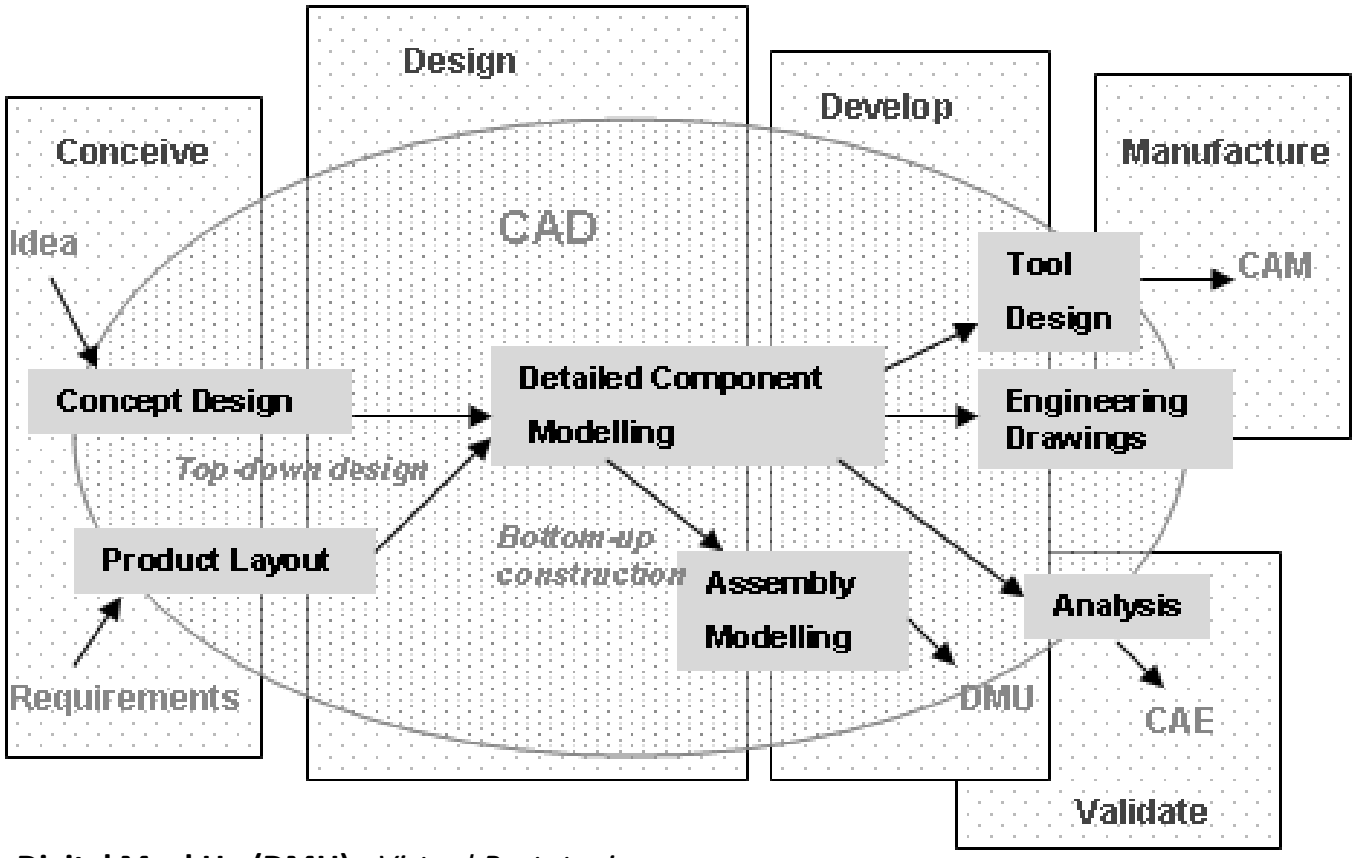
組長 機構 控制 預算 加工 報告 .....



Engineering Design Process



CAD/CAE/CAM



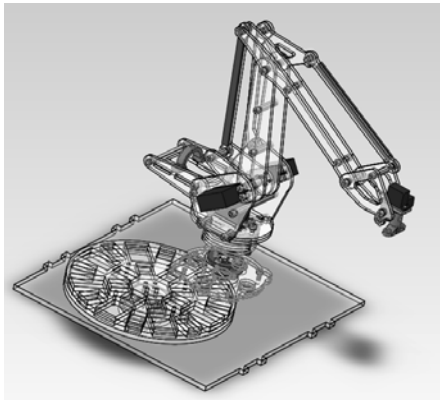
Digital MockUp (DMU): *Virtual Prototyping*.

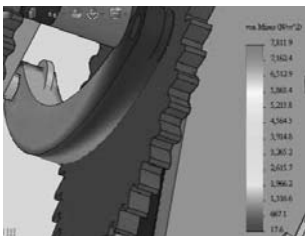
Common **CAD/CAE** Packages

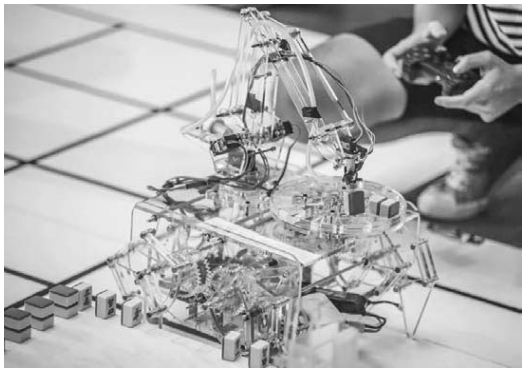
- Geometric Modeling
  - AutoCAD
  - Pro-Engineer
  - I-DEAS
  - CATIA
  - Solid Edge
  - SolidWorks
- Finite-Element Modeling
  - ANSYS
  - ABAQUS
  - MSC PATRAN/NASTRAN
  - MARC
  - COSMOS
  - Pro-Mechanica
  - LS-DYNA

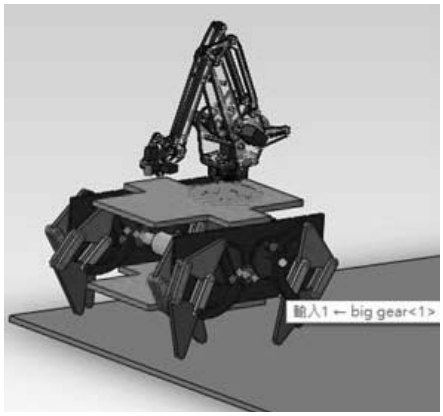
- **CAM**: CNC, 3D Printer, 雕刻機, 雷射切割機……

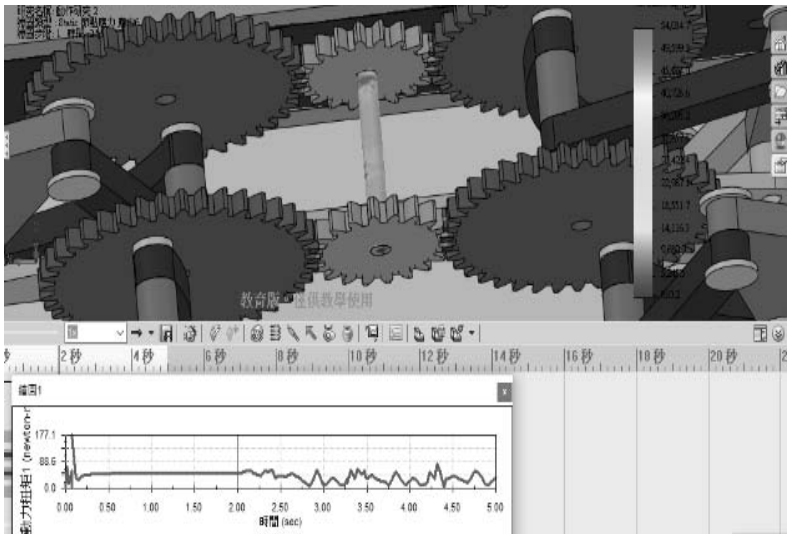
Example











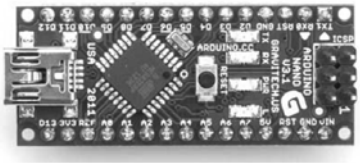
系統控制

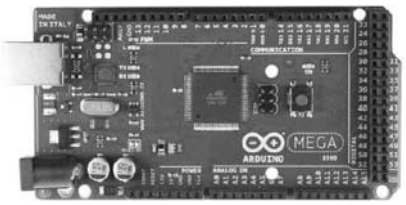
Arduino IDE 1.8.13

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

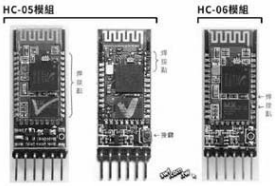
Ex: Arduino:

1. Arduino UNO R3 ($100~$800)
2. Arduino NANO ($70~$300)
3. Arduino MEGA 2560 ($300~$1000)
4. Arduino Due ($600~$2000)



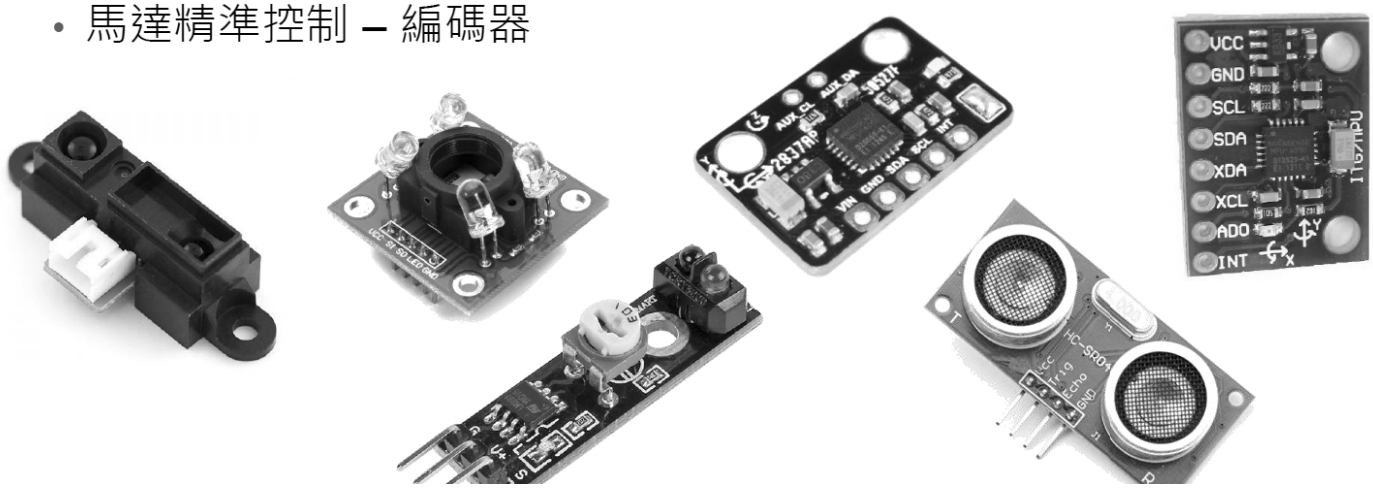


- Servo motor:
  Tower Pro SG90($45~$150)
- DC motor
- Bluetooth:
  1. HC-05
  2. HC-06($100~$250)
- Android smartphone
- Laptop



- Arduino 的軟體開發環境是開放源碼，可以在Arduino官網免費下載，它所用的程式語言語法類似於 C/C++，具備文字編輯介面、常用工具欄、圖形化控制介面及錯誤編輯器，運用這個工具進行程式的上傳，及各個感應器的溝通。

- 感測器
  - 簡易顏色、障礙物判別 – 紅外線感測器
  - 位置、加速度、角速度感測 – 陀螺儀、加速規(MPU6050)
  - 精準顏色判別 – 顏色感測器(TCS 3200M)
  - 障礙物判別、距離感測 – 超音波測距模組(HC SR40)
  - 馬達精準控制 – 編碼器

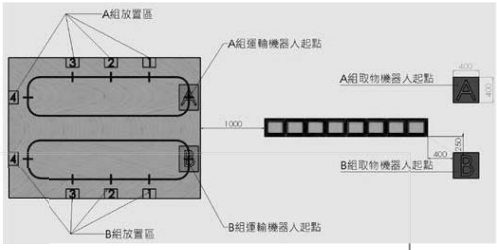


Project Management Triangle

Project Management

Scope

製作機器人完成特定任務 +



1. 取物機器人：
   (1) 機器人之尺寸須小於 長 40 公分、寬 40 公分，高度不限。
   (2) 使用無線遙控的方式操作。
   (3) 取物機器人須為足型，不可使任何非足型部位接觸地面 (不可使用任何輪子與支撐結構輔助移動，禁止單純支撐結構碰觸地面)。
   (4) 禁止購買與使用現成機器人系統，但零組件、控制板、驅動器、感測器與遙控器等，可以使用市售之相關產品。
   (5) 競賽時，每組機器人上皆須標記明顯的班級與組別，如甲班第一組則以「甲-1」為編號，以此類推。

系上將提供經費 10000 元予每組製作機器人系統 Cost

Today: 第2週 Time

時間	第 1 週	第 6 週	第 12 週	第 17 週	第 18 週
工作	完成分組	專題計劃書	期中報告	實作競賽	期末報告
內容	概念設計、細部規劃、零件圖及組合圖、加工規劃書、系統分析、經費規劃		分組實作		專題實作展

Project Management Tool: MS Project

Microsoft Project - Sample Project.mpp

File Edit View Insert Format Tools Project Window Help

Project Summary

	WBS	Task Name	Cost
1	1	Project Summary	$38,000.00
2	1.1	Design Phase	$18,400.00
3	1.1.1	First Design Phase	$4,000.00
4	1.1.1.1	Start Milestone	$0.00
5	1.1.1.2	Design Task 1	$4,000.00
6	1.1.2	Second Design Phase	$14,400.00
7	1.1.2.1	Design Task 2	$6,000.00
8	1.1.2.2	Design Task 3	$4,000.00
9	1.1.2.3	Design Task 4	$4,400.00
10	1.1.2.4	End Design Milestone	$0.00
11	1.2	Programming Phase	$10,000.00
12	1.2.1	Program Task 1	$6,000.00
13	1.2.2	Program Task 2	$4,000.00
14	1.2.3	End Program Milestone	$0.00
15	1.3	Testing Phase	$9,600.00
16	1.3.1	Test Task 1	$2,400.00

Gantt Chart

January February

12/31 01/07 01/14 01/21 01/28 02/04 02/11 02/18 02/25

01/01 02/09 02/23

Ready

Project Management Tool: MS Project

WBS Chart Pro - [Sample Project.wbs]

File Edit View Format Tools Window Help

Design Task 1

Critical Path View 2

1 Project Summary 720 hours $38,000.00

1.1 Design Phase 360 hours $18,400.00

1.2 Programming Phase 200 hours $10,000.00

1.3 Testing Phase 160 hours $9,600.00

1.1.1 First Design Phase 80 hours $4,000.00

1.1.1.1 Start Milestone 0 hours $0.00

1.1.1.2 Design Task 1 80 hours $4,000.00

1.1.2 Second Design Phase 280 hours $14,400.00

1.1.2.1 Design Task 2 120 hours $6,000.00

1.1.2.2 Design Task 3 80 hours $4,000.00

1.1.2.3 Design Task 4 80 hours $4,400.00

1.2.1 Program Task 1 120 hours $6,000.00

1.2.2 Program Task 2 80 hours $4,000.00

1.2.3 End Program Milestone 0 hours $0.00

1.3.1 Test Task 1 40 hours $2,400.00

1.3.2 Test Task 2 40 hours $2,400.00

1.3.3 Test Task 3 80 hours $4,800.00

1.3.4 End Milestone 0 hours $0.00

學術文獻搜尋

電子資源查詢

服務諮詢：資料庫：資訊服務組 | 06-2757575 轉 65780 |
電子期刊：期刊組 | 06-2757575 轉 65736 |
醫學類資源：醫分館 | 06-2353535 轉 5121 |

歡迎使用成功大學圖書館電子資源，請依據您的需要選擇合適系統進行查詢。本校師生自校外使用相關系統不需設定代理伺服器，但如您遇無法使用或順利取得全文狀況時，請將發生之狀況告知我們，以協助及早排除。

- 電子資源查詢系統

新版電子資源查詢系統，整合了總圖及醫分館之資料庫、電子期刊、講習課程等相關資訊，單一介面操作簡單，並可針對個人常用資源進行個人化設定。

- 電子資源整合查詢

利用簡單的關鍵字搜尋，同時查詢多種由本館精選重要之期刊論文、博碩士論文、專利標準、電子書、報紙新聞等不同的資料庫系統；若要查詢單一資料庫或電子期刊，請點選資料庫或電子期刊查詢系統。

資料庫/ 電子期刊

- Web of Science
- Science Direct
- ASME
- IEEE
- ……

成功大學圖書館 Lib.NCKU 電子資源查詢系統 E-Resources Gateway

資料庫 | 電子期刊 | 電子書 | 醫學資源 | 書目管理軟體 | 講習課程 |

【資源查詢】

資源名稱 (tim) | 所有資源 | 精確檢索 | 查詢 | 進階查詢

最新消息

日期	
2015/01/28	電子資源使用規範宣導
2015/01/14	PNAS OA期刊投稿優惠
2015/01/14	BioMed Central OA 期刊投稿優惠
2015/01/05	2015年新、停訂資料庫清單
2015/01/01	新增/試用資料庫及推廣活動消息

成功大學 | 圖書館 | English

成功大學圖書館 Lib.NCKU 電子資源管理系統 E-Resources Gateway

使用者登入

帳號：

密碼：

登入

系統說明

本系統提供讀者在單一檢索介面，查詢、瀏覽及連結使用多元化的館藏資料庫、電子書、電子期刊及網路資源，並提供線上個人化服務，便捷讀者使用電子資源。

認證方式：

帳號密碼與「館藏查詢系統」個人化服務之帳號密碼相同。

(1) 帳號：教職員證或學生證之證號。

(2) 密碼：「館藏查詢系統」個人化服務之密碼。

忘記密碼或要更改密碼，詳見成功入口

本校附設醫院同仁

(1) 帳號:識別證號前加4碼0

(2) 密碼預設值:【身分證】後4碼+【生日】後4碼

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legged walking robot

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- Conference Publications (66)
- Journals & Magazines (3)

PUBLICATION YEAR

Single Year | Range

1985 — 2014

From: 1985

To: 2014

AUTHOR

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Results per page 25 | Sort by: Newest First

Select All on Page | Deselect All | « First | 1 2 3 Last »

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LAURON V: A versatile six-legged walking robot with advanced maneuverability

Roennau, A. ; Heppner, G. ; Nowicki, M. ; Dillmann, R.
Advanced Intelligent Mechatronics (AIM), 2014 IEEE/ASME International Conference on
DOI: 10.1109/AIM.2014.6878051
Publication Year: 2014 , Page(s): 82 - 87
Cited by: Papers (1)
IEEE CONFERENCE PUBLICATIONS

Multimedia

| © | Quick Abstract | PDF (3572 KB)

SEARCH HISTORY

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STANDARDS DICTIONARY TERMS (What's this?)

Browse Standards Dictionary

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IEEE

2014 IEEE/ASME International Conference on
Advanced Intelligent Mechatronics (AIM)
Besançon, France, July 8-11, 2014

LAURON V: A Versatile Six-Legged Walking Robot with Advanced Maneuverability

A. Roennau[1], G. Heppner[1], M. Nowicki[2] and R. Dillmann[1]

***Abstract*— Adaptive multi-legged walking robots are predestined to be applied in rough and hazardous terrain. Their walking and climbing skills allow them to operate at places that are unreachable for most wheeled vehicles. In this paper, we present the design and development of the new six-legged walking robot LAURON V with its improved kinematics and robust mechanical structure. Each leg has four independent joints that enable LAURON to cope with steep inclines, large obstacles and makes it possible to manipulate objects with its front legs. Autonomy, robustness and a large payload capacity together with its impressive terrain adaptability make LAURON V highly suitable for all kinds of field applications.**

I. Introduction

Walking robots, especially hexapods, are designed to walk in difficult, rough and often unknown terrain. Stable locomotion in such an environment, however, is not easy and therefore many researcher around the world continuously develop new concepts to improve a variety of legged robots.

The FZI Research Center for Information Technology has a long standing tradition of walking robots [1], [2]. In particular the LAURON hexapods [3], [4], [5], [6] reach back over 20 years. With each iteration this biologically inspired hexapod was made more robust, learned new skills and continued to extend the previous research results. Of course, there are other walking robots that show advanced walking skills like HyQ [7] or LS3 [8]. Many six-legged walking robots apply the predominant yaw-pitch-pitch kinematics, which is based on the typical insect-like leg design with three joints: Messor II [9], AMOS-WD06[10], BILL-Stick[11], DLR Crawler [12] and Anton [13]. The impressive number of 18 degrees of freedom (DoF) offers a good flexibility and walking abilities on mainly flat terrain with obstacles. However, if the terrain becomes more challenging, as it is the case with larger obstacles or steep slopes, this kinematic design reaches its limit as the orientation of the feet cannot be chosen freely. Some of these limitations are addressed by sophisticated motion planning methods and foothold selection approaches [14],[15],[16]. But of course, these approaches require detailed environmental information and an exact robot localization. Certainly, there are also many multi-legged walking robots with an advanced leg design (more than three degrees of freedom per leg): CR200 [17], Athlete [18], LEMUR [19] and CREX [20]. Their additional degrees of freedom (DoF) are mostly designed for special tasks like object manipulation and are rarely used to improve their climbing or walking skills. For example, object manipulation with legged robots has been demonstrated by Boston Dynamics' BigDog [21] and by six-legged robots like CREX [20] or LEMUR [19]. In spite of the continuous improvement of LAURON IVc we reached the limits of its kinematic design and hardware, provoking development of LAURON V. The main design objectives were the kinematic improvements, enabling LAURON V to walk over steeper inclines and larger obstacles as well as to enable manipulation tasks. Some parts, like the main body have been redesigned while proven mechanical designs were sustained. With minor changes, the behavior-based control was adapted to the new hardware, allowing LAURON V to inherit all of LAURON IVc's capabilities. A new high power computer system, a modular sensor setup and an enhanced electronics structure complete the redesign, improving its autonomy. LAURON V's size, mechatronic robustness, payload capacity and maneuverability make it absolutely unique and address the requirements of challenging field missions like search and rescue operations as well as inspection, exploration and surveillance tasks (see Fig. 1).

Fig. 1: Walking Robot LAURON V in the Field.

Section 2 of this work will present a short history and the design objectives for LAURON V. Section 3 and 4 focus on kinematics and mechanical design. The hierarchical control architecture will be described in Section 5. LAURON V's advanced skills and the evaluation at the EU Taranis 2013 are content of the Sections 6 and 7. Finally, Section 8 will conclude our work and give a short outlook to future work.

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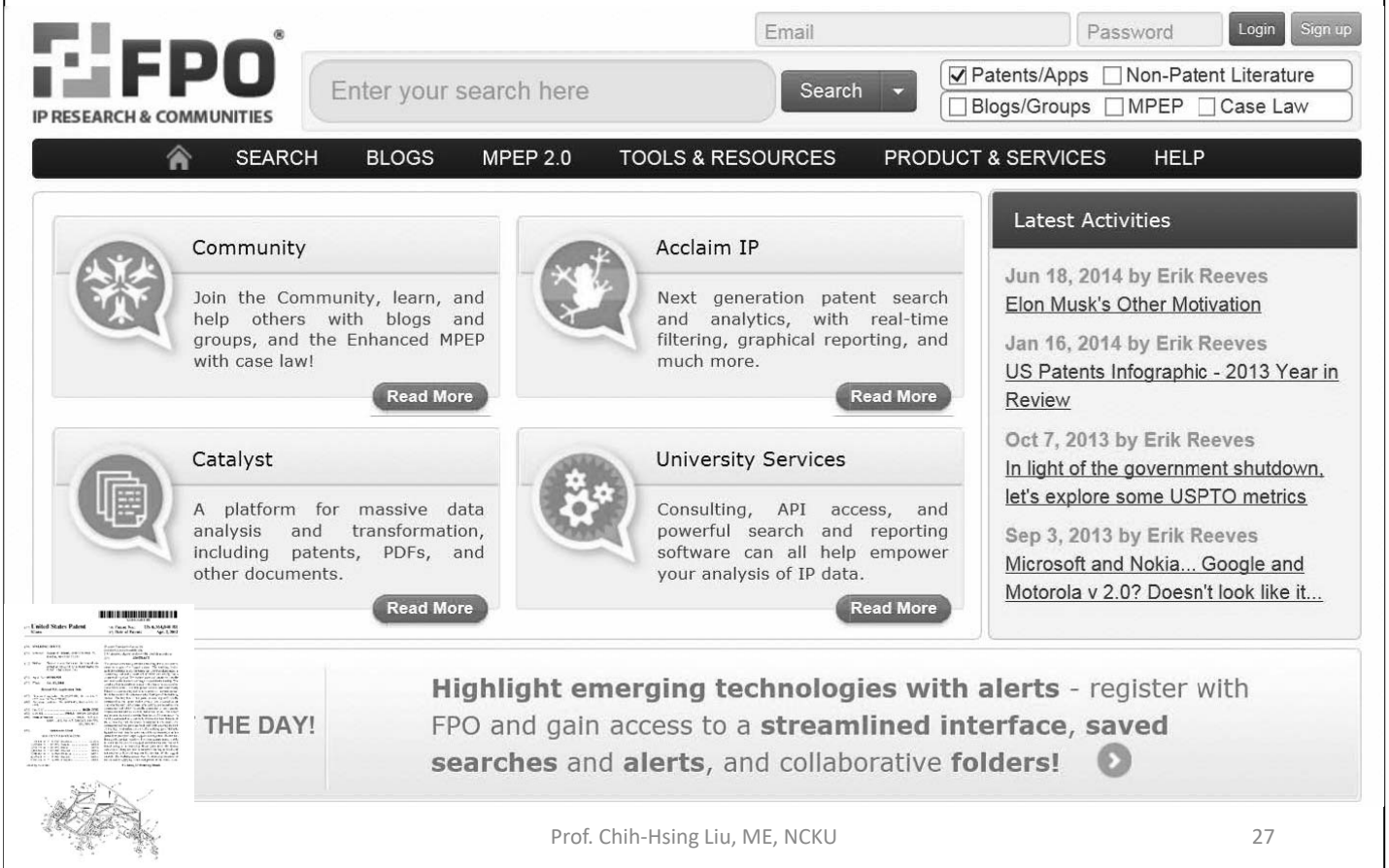

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